**DATABASE MANAGEMENT SYSTEMS (IT 2040)**

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**Question 1**

What is a view in a database?

- A) A physical table stored permanently on disk
- B) A virtual table derived from base tables that doesn't necessarily exist in physical form
- C) A backup copy of a table
- D) An index on a table

**Answer: B**

---

**Question 2**

Which statement about views is FALSE?

- A) Views can simplify complex queries
- B) Views can provide security by restricting data access
- C) All views are updateable
- D) Views don't physically store data **Answer: C**.

---

**Question 3**

Given a view created as:

CREATE VIEW dept_summary AS

SELECT deptId, COUNT(*) as empCount, AVG(salary) as avgSalary

FROM Employee

GROUP BY deptId;

Can you execute: UPDATE dept_summary SET avgSalary = 50000 WHERE deptId = 10;?

- A) Yes, it updates the average salary
- B) No, views with aggregate functions are not updateable
- C) Yes, but it updates all employee salaries
- D) Yes, but only if there's an INSTEAD OF trigger

**Answer: B (or D if trigger exists)**

---

**Question 4**

What happens when you drop a base table that a view depends on?

- A) The view is automatically dropped
- B) The view remains but queries fail
- C) The view is converted to a regular table
- D) Depends on the DBMS

**Answer: D**

---

**Question 5**

In T-SQL, which is the correct way to declare a variable?

- A) VAR @empId INT;
- B) DECLARE @empId INT;
- C) SET @empId INT;
- D) INT @empId;

**Answer: B**

---

**Question 6**

What is the initial value of a declared T-SQL variable?

- A) 0 for numbers, empty string for text
- B) NULL
- C) Undefined
- D) Must be initialized immediately

**Answer: B**

---

**Question 7**

How do you assign a value to a variable in T-SQL?

- A) @empId = 100;
- B) SET @empId = 100;
- C) LET @empId = 100;
- D) ASSIGN @empId = 100;

**Answer: B**

---

**Question 8**

What does the BREAK statement do in a WHILE loop?

- A) Skips current iteration and continues to next
- B) Exits the innermost WHILE loop completely
- C) Pauses execution temporarily
- D) Exits all nested loops

**Answer: B**

---

**Question 9**

What does the CONTINUE statement do in a WHILE loop?

- A) Exits the loop
- B) Skips remaining statements in current iteration and restarts the loop
- C) Continues to the next loop
- D) Does nothing

**Answer: B**

---

**Question 10**

What is the key difference between a stored function and a stored procedure?

- A) Functions can have parameters, procedures cannot
- B) Functions must return a single value, procedures may return multiple values via OUTPUT
- C) Procedures execute faster
- D) No significant difference

**Answer: B**

---

**Question 11**

Can a stored function use OUTPUT parameters?

- A) Yes, just like procedures
- B) No, functions only use IN parameters and RETURN value
- C) Only for table-valued functions
- D) Yes, but only one OUTPUT parameter

**Answer: B**

---

**Question 12**

How do you call a stored function in T-SQL?

- A) EXEC functionName(parameters);
- B) CALL functionName(parameters);
- C) SELECT dbo.functionName(parameters);
- D) EXECUTE FUNCTION functionName(parameters); **Answer: C**

---

**Question 13** Given:

CREATE FUNCTION dbo.getBonus(@salary FLOAT) RETURNS FLOAT

AS

BEGIN

RETURN @salary * 0.10

END

Which call is INCORRECT?

- A) SELECT dbo.getBonus(50000);
- B) SELECT dbo.getBonus(salary) FROM Employee;
- C) EXEC dbo.getBonus 50000;
- D) SELECT *, dbo.getBonus(salary) as bonus FROM Employee; **Answer: C**

---

**Question 14**

What is the default parameter mode for stored procedures if not specified?

- A) OUTPUT
- B) IN
- C) INOUT
- D) Must always be specified

**Answer: B**

---

**Question 15**

Given a procedure:

CREATE PROCEDURE calcStats(@deptId INT, @maxSal FLOAT OUTPUT, @minSal FLOAT OUTPUT)

```
AS
BEGIN
   SELECT @maxSal=MAX(salary), @minSal=MIN(salary)
   FROM Employee WHERE deptId=@deptId
END
```

Which call is CORRECT?

- A) EXEC calcStats 10, @max, @min;
- B) DECLARE @max FLOAT, @min FLOAT; EXEC calcStats 10, @max OUTPUT, @min OUTPUT;
- C) SELECT calcStats(10, @max, @min);
- D) CALL calcStats(10, @max, @min);

**Answer: B**

---

**Question 16**

What is a trigger in a database?

- A) A scheduled task that runs periodically
- B) A special stored procedure that automatically executes in response to DML/DDL events
- C) A constraint on a table
- D) A backup mechanism

**Answer: B**

---

**Question 17**

What are the INSERTED and DELETED virtual tables in triggers?

- A) Permanent system tables
- B) Temporary tables created during trigger execution storing affected rows

- C) Backup tables
- D) User-created tables

**Answer: B**

---

**Question 18**

For an INSERT operation, what data appears in trigger tables?

- A) INSERTED has new rows, DELETED is empty  B) DELETED has new rows, INSERTED is empty
- C) Both tables have the new rows
- D) Both tables are empty **Answer: A**

---

**Question 19**

For an UPDATE operation, what appears in INSERTED and DELETED tables?

- A) INSERTED has old values, DELETED has new values  B) INSERTED has new values, DELETED has old values
- C) Both have new values
- D) Only INSERTED has values

**Answer: B**

---

**Question 20**

For a DELETE operation, what appears in trigger tables?

- A) INSERTED has deleted rows, DELETED is empty  B) DELETED has deleted rows, INSERTED is empty
- C) Both tables have deleted rows
- D) Both tables are empty

**Answer: B**

---

**Question 21**

What is the difference between AFTER and INSTEAD OF triggers?

- A) No difference, just different syntax
- B) AFTER executes after DML completes, INSTEAD OF executes in place of DML
- C) AFTER is faster

- D) INSTEAD OF is only for DDL

**Answer: B**

---

**Question 22**

Can AFTER triggers be defined on views?

- A) Yes, on any view
- B) No, AFTER triggers cannot be defined on views
- C) Only on updateable views
- D) Only with special permissions **Answer: B**

---

**Question 23**

How many INSTEAD OF triggers can be defined per INSERT operation on a table?

- A) Unlimited
- B) At most one
- C) Exactly two (before and after)
- D) None - INSTEAD OF only for views

**Answer: B**

---

**Question 24**

Given a trigger:

```
CREATE TRIGGER prevent_delete
ON Employee
INSTEAD OF DELETE
AS
BEGIN
   PRINT 'Deletion not allowed'
END
```

What happens when you execute DELETE FROM Employee WHERE eid = 100;?

- A) Employee 100 is deleted
- B) Message printed, but employee is NOT deleted
- C) Error occurs
- D) All employees are deleted

**Answer: B**

---

**Question 25**

What does ROLLBACK TRANSACTION do in a trigger?

- A) Saves changes permanently
- B) Undoes the entire transaction including the triggering operation
- C) Only undoes trigger code
- D) Creates a backup **Answer: B**

---

**Question 26**

Which trigger is better for implementing validation logic?

- A) AFTER trigger with ROLLBACK if validation fails
- B) INSTEAD OF trigger that only executes DML if validation passes
- C) Both are equally good
- D) Neither - use CHECK constraints

**Answer: B**

---

**Question 27**

Can triggers call other stored procedures?

- A) No, triggers are isolated
- B) Yes, triggers can execute procedures
- C) Only AFTER triggers can
- D) Only with special permissions

**Answer: B**

---

**Question 28**

What happens if a trigger raises an error?

- A) Error is ignored, operation continues
- B) Trigger code stops, but DML operation completes
- C) Entire transaction is rolled back (for AFTER triggers)
- D) Only the trigger is rolled back

**Answer: C**

---

**Question 29**

Which is a valid advantage of using views?

- A) Faster query execution
- B) Security - restrict user access to specific columns/rows
- C) Automatic backup
- D) Increase storage space **Answer: B**

---

**Question 30**

What is a disadvantage of views?

- A) Too much disk space usage
- B) Cannot query multiple tables
- C) Performance overhead - queries on views must be translated to base table queries
- D) Limited to one view per table

**Answer: C**

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**SECTION B: MULTIPLE ANSWER QUESTIONS (Select ALL that apply) - 10 Questions**

**Question 31**

Which types of views are generally NOT updateable? **(Select all that apply)**

- A) Views with aggregate functions (SUM, AVG, COUNT)
- B) Views joining multiple tables
- C) Views with DISTINCT
- D) Simple SELECT from single table
- E) Views with GROUP BY
- F) Views with subqueries in SELECT

**Answers: A, B, C, E, F**

---

**Question 32**

Which statements about T-SQL variables are TRUE? **(Select all that apply)**

- A) Variable names must start with @
- B) Variables must be declared before use
- C) Variables are initialized to NULL by default

- D) Variables can store query results
- E) Variables persist across sessions
- F) Can use variables in WHERE clauses **Answers: A, B, C, D, F**

---

**Question 33**

Which control-of-flow statements are available in T-SQL? **(Select all that apply)**

- A) IF...ELSE
- B) WHILE
- C) FOR loop
- D) BREAK
- E) CONTINUE
- F) SWITCH...CASE

**Answers: A, B, D, E**

---

**Question 34**

Advantages of stored procedures include: **(Select all that apply)**

- A) Reduced network traffic (logic executes on server)
- B) Improved performance (pre-compiled execution plan)
- C) Enhanced security (users don't need direct table access)
- D) Code reusability across applications
- E) Automatic data backup
- F) Centralized business logic

**Answers: A, B, C, D, F**

---

**Question 35**

Which are valid characteristics of stored functions? **(Select all that apply)**

- A) Must return a value
- B) Can use OUTPUT parameters
- C) Can be called in SELECT statements
- D) Can modify database tables
- E) Parameters are input-only (IN mode)
- F) Can call other functions

**Answers: A, C, E, F**

---

**Question 36**

What can trigger a DML trigger? **(Select all that apply)**

- A) INSERT statement
- B) UPDATE statement  C) DELETE statement  D) SELECT statement
- E) TRUNCATE statement
- F) MERGE statement

**Answers: A, B, C, F**

---

**Question 37**

For a trigger on Employee table with UPDATE operation, INSERTED and DELETED tables contain: **(Select all that apply)**

- A) INSERTED: new values after update
- B) DELETED: old values before update
- C) Both tables have same structure as Employee table
- D) INSERTED: only changed columns
- E) DELETED: only for deleted rows
- F) Both tables can have multiple rows (if multiple rows updated) **Answers: A, B, C, F**

---

**Question 38**

When should you use INSTEAD OF triggers? **(Select all that apply)**

- A) To update views that join multiple tables
- B) To implement custom INSERT logic
- C) To prevent certain operations entirely
- D) To audit changes after they occur
- E) To override default DML behavior
- F) When AFTER triggers are not sufficient

**Answers: A, B, C, E, F**

---

**Question 39**

Which scenarios require stored procedures over functions? **(Select all that apply)**

- A) Need to return multiple values
- B) Need to modify database tables
- C) Need to use OUTPUT parameters
- D) Need to call from SELECT statement
- E) Need to handle transactions
- F) Need to execute dynamic SQL

**Answers: A, B, C, E, F**

---

**Question 40**

Best practices for triggers include: **(Select all that apply)**

- A) Keep trigger logic simple and fast
- B) Avoid lengthy transactions in triggers
- C) Use triggers for complex business logic
- D) Document what triggers do
- E) Consider using procedures called from triggers for complex logic  F) Avoid recursive triggers

**Answers: A, B, D, E, F**

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**SECTION C: ADVANCED SCENARIO-BASED QUESTIONS - 10 Questions**

**Question 41**

**Scenario:** Banking system with Account(accountNo, balance), Transaction(transId, accountNo, amount, type, date).

**Requirement:** Automatically update account balance when transactions are inserted.

**Which trigger is MOST robust?**

```
-- Option A
CREATE TRIGGER update_balance
ON Transaction
AFTER INSERT
AS
BEGIN
```

```sql
    UPDATE Account
    SET balance = balance +
        (SELECT SUM(CASE WHEN type='credit' THEN amount ELSE -amount END)
         FROM inserted WHERE accountNo = Account.accountNo)
    WHERE accountNo IN (SELECT accountNo FROM inserted)
END

-- Option B
CREATE TRIGGER update_balance
ON Transaction
AFTER INSERT
AS
BEGIN
    UPDATE a
    SET a.balance = a.balance +
        CASE WHEN i.type='credit' THEN i.amount ELSE -i.amount END
    FROM Account a
    INNER JOIN inserted i ON a.accountNo = i.accountNo
END
```

**Answer: B**

---

**Question 42**

**Scenario:** Employee(eid, name, salary, managerId) with self-referencing managerId.

**Requirement:** Ensure employee salary never exceeds manager's salary.

**Which trigger handles UPDATE correctly?**

```sql
-- Option A
CREATE TRIGGER check_salary
ON Employee
AFTER UPDATE
AS
BEGIN
    IF EXISTS (
```

```
    SELECT * FROM inserted i
    JOIN Employee m ON i.managerId = m.eid
    WHERE i.salary > m.salary
  )
  BEGIN
    ROLLBACK TRANSACTION
    RAISERROR('Salary exceeds manager', 16, 1)
  END
END

-- Option B
CREATE TRIGGER check_salary
ON Employee
AFTER UPDATE
AS
BEGIN
  IF UPDATE(salary) OR UPDATE(managerId)
  BEGIN
    IF EXISTS (
      SELECT * FROM inserted i
      JOIN Employee m ON i.managerId = m.eid
      WHERE i.salary > m.salary AND i.managerId IS NOT NULL
    )
    BEGIN
      ROLLBACK TRANSACTION
      RAISERROR('Salary exceeds manager', 16, 1)
    END
  END
END
```

**Answer: B**

---

**Question 43**

**Scenario:** Library: Book(isbn, title), Copy(copyId, isbn), Loan(loanId, copyId, memberId, loanDate, returnDate).

**Create view showing books currently on loan:**

```
CREATE VIEW books_on_loan AS
SELECT b.isbn, b.title, c.copyId, l.memberId, l.loanDate
FROM Book b
JOIN Copy c ON b.isbn = c.isbn
JOIN Loan l ON c.copyId = l.copyId
WHERE l.returnDate IS NULL;
```

**Can you execute: UPDATE books_on_loan SET title = 'New Title' WHERE isbn = '12345';?**

- A) Yes, updates the book title
- B) No, view joins multiple tables
- C) Yes, with INSTEAD OF trigger
- D) Both B and C

**Answer: D**

---

**Question 44**

**Scenario:** Create function to calculate employee's total work percentage across all departments.

**Given:** Employee(eid, name), Works(eid, deptId, pct_time) **Which**

**function is correct?**

```
-- Option A
CREATE FUNCTION dbo.getTotalPct(@empId INT) RETURNS FLOAT
AS
BEGIN
   DECLARE @total FLOAT
   SELECT @total = SUM(pct_time)
   FROM Works
   WHERE eid = @empId
   RETURN @total
END


-- Option B
```

```
CREATE FUNCTION dbo.getTotalPct(@empId INT) RETURNS FLOAT
AS
BEGIN
   DECLARE @total FLOAT
   SELECT @total = ISNULL(SUM(pct_time), 0)
   FROM Works
   WHERE eid = @empId
   RETURN @total
END
```

**Answer: B**

---

**Question 45**

**Scenario:** Product(pid, name, price, stock), Sale(saleId, pid, quantity, saleDate).

**Requirement:** Prevent sales when stock insufficient, and reduce stock when sale succeeds.

**Best trigger design:**

```
CREATE TRIGGER process_sale
ON Sale
INSTEAD OF INSERT
AS
BEGIN
   -- Check stock sufficient
   IF EXISTS (
      SELECT * FROM inserted i
      JOIN Product p ON i.pid = p.pid
      WHERE i.quantity > p.stock
   )
   BEGIN
      RAISERROR('Insufficient stock', 16, 1)
      RETURN
   END

   -- Insert sale
```

```
    INSERT INTO Sale SELECT * FROM inserted

    -- Reduce stock
    UPDATE p
    SET p.stock = p.stock - i.quantity
    FROM Product p
    JOIN inserted i ON p.pid = i.pid
END
```

**Why use INSTEAD OF instead of AFTER?**

- A) AFTER cannot access INSERTED table
- B) INSTEAD OF allows checking before operation and combining logic
- C) AFTER is slower
- D) INSTEAD OF is required for stock updates **Answer: B**

---

**Question 46**

**Scenario:** HR system - Employee(eid, name, salary, deptId), Department(deptId, name, budget).

**Create procedure to give raise to all employees in a department, ensuring total salary doesn't exceed budget:**

```
CREATE PROCEDURE giveRaise(
    @deptId INT,
    @raisePercent FLOAT,
    @success BIT OUTPUT
)
AS
BEGIN
    DECLARE @totalSalary FLOAT, @budget FLOAT, @newTotal FLOAT

    SELECT @totalSalary = SUM(salary)
    FROM Employee WHERE deptId = @deptId

    SELECT @budget = budget FROM Department WHERE deptId = @deptId
```

```
    SET @newTotal = @totalSalary * (1 + @raisePercent/100)

    IF @newTotal <= @budget
    BEGIN
        UPDATE Employee
        SET salary = salary * (1 + @raisePercent/100)
        WHERE deptId = @deptId
        SET @success = 1
    END
    ELSE
        SET @success = 0
END
```

**To call this procedure:**

```
DECLARE @result BIT
EXEC giveRaise 10, 5.0, @result OUTPUT
IF @result = 1
    PRINT 'Raise applied'
ELSE
    PRINT 'Raise would exceed budget'
```

**This pattern demonstrates:**

- A) Transaction management
- B) Business rule enforcement in stored procedures
- C) OUTPUT parameter usage
- D) All of the above

**Answer: D**

---

**Question 47**

**Scenario:** Audit system tracks all changes to sensitive Employee salary field.

**Tables:** Employee(eid, name, salary), SalaryAudit(auditId IDENTITY, eid, oldSalary, newSalary, changeDate, changedBy)

**Create comprehensive audit trigger:**

```
CREATE TRIGGER audit_salary
ON Employee
```

```
AFTER UPDATE
AS
BEGIN
   IF UPDATE(salary)
   BEGIN
      INSERT INTO SalaryAudit (eid, oldSalary, newSalary, changeDate, changedBy)
SELECT
         i.eid,
         d.salary,
         i.salary,
         GETDATE(),
         SYSTEM_USER
      FROM inserted i
      JOIN deleted d ON i.eid = d.eid
      WHERE i.salary <> d.salary
   END
END
```

**What does UPDATE(salary) check?**

- A) If salary is NULL
- B) If salary column was included in UPDATE statement (even if value unchanged)
- C) If salary value actually changed
- D) If salary is valid

**Answer: B**

---

**Question 48**

**Scenario:** Student registration system prevents overbooking courses.

**Tables:** Course(courseId, title, maxStudents), Enrollment(studentId, courseId, enrollDate)

**Create trigger to enforce maximum:**

```
CREATE TRIGGER prevent_overbook
ON Enrollment
INSTEAD OF INSERT
AS
```

```
BEGIN
   DECLARE @courseId INT, @maxStudents INT, @currentCount INT

   SELECT @courseId = courseId FROM inserted
   SELECT @maxStudents = maxStudents FROM Course WHERE courseId = @courseId
   SELECT @currentCount = COUNT(*) FROM Enrollment WHERE courseId = @courseId

   IF @currentCount < @maxStudents
      INSERT INTO Enrollment SELECT * FROM inserted
   ELSE
      RAISERROR('Course is full', 16, 1)
END
```

**What's the flaw in this trigger?**

- A) Should use AFTER instead of INSTEAD OF
- B) Doesn't handle multiple students enrolling simultaneously
- C) Should check maxStudents before counting
- D) Missing transaction handling

**Answer: B**

---

**Question 49**

**Scenario:** Complex view joins Student, Enrollment, and Course tables. User tries to update course name through the view.

**What happens?**

```
CREATE VIEW student_courses AS
SELECT s.sid, s.name AS studentName, c.courseId, c.name AS courseName
FROM Student s
JOIN Enrollment e ON s.sid = e.sid
JOIN Course c ON e.courseId = c.courseId;

-- User executes:
UPDATE student_courses SET courseName = 'Database Systems' WHERE courseId =
'CS101';
```

**Result:**

- A) Update succeeds and course name changes
- B) Update fails - views with joins are not directly updateable
- C) Update succeeds but only for one student's enrollment
- D) Update succeeds with INSTEAD OF trigger

**Answer: B**

**Question 50**

**Scenario**: HR database: Employee(eid, name, salary, deptId), Department(deptId, deptName, budget).

**Requirement**: Create a function to calculate what percentage of department budget is spent on salaries.

**Correct function:**

```
CREATE FUNCTION dbo.getSalaryPercentage(@deptId INT)
RETURNS FLOAT
AS
BEGIN
    DECLARE @totalSalary FLOAT, @budget FLOAT, @percentage FLOAT

    SELECT @totalSalary = SUM(salary)
    FROM Employee
    WHERE deptId = @deptId

    SELECT @budget = budget
    FROM Department
    WHERE deptId = @deptId

    IF @budget > 0
        SET @percentage = (@totalSalary / @budget) * 100
    ELSE
        SET @percentage = 0
```

RETURN @percentage

END

This function is called using:

- A) EXEC dbo.getSalaryPercentage 101
- B) SELECT dbo.getSalaryPercentage(101)
- C) CALL getSalaryPercentage(101)
- D) EXECUTE FUNCTION getSalaryPercentage(101)

**Answer: B**

---

**Question 51**

Which of the following statements are true related to constraints? (Select one or more)

• A) If a subtype participates in a relationship that is the same as the other subtypes, that relationship could be added to the supertype.

• B) When an entity belongs to only one subtype in the hierarchy, the relationship is total and disjoint.

• C) A bowler and batsman (wicketkeeper is also a batsman) which are subtypes of a cricketer type is total and disjoint.

• D) When an entity instance may be a member of multiple subtypes, or does not have to be a member of a subtype, the specialization is overlapping and total.

**Answer: A**

---

**Question 52**

Consider the following table definition:

CREATE TABLE Product (

pid CHAR(4) PRIMARY KEY,

pname VARCHAR(30),

manufactureDate DATETIME DEFAULT GETDATE(),

qty INT,

CONSTRAINT pid_chk CHECK (pid LIKE 'PSTO-510-5[0-5]-577'), CONSTRAINT qty_chk CHECK (qty >= 0)

);

Which of the following statements are true related to the above definition? (Select one or more)

- A) Consider a product with a quantity (qty) of 10. An update to deduct 15 from the available quantity in the above table will not be successful.
- B) "T12" is a valid pid that could be stored in the Product table.
- C) "P333" is a valid pid that could be stored in the Product table.
- D) Executing the following insert statement will cause an error: INSERT INTO Product(pid, qty) VALUES ('T000', 50)
- E) After executing the following insert statement in the above table, the row inserted will have two null values: INSERT INTO Product(pid, qty) VALUES ('T000', 50)

**Answer: A, D**

---

**Question 53**

Consider the following table definition:

CREATE TABLE Product (

pid CHAR(4) PRIMARY KEY,

pname VARCHAR(30),

manufactureDate DATETIME DEFAULT GETDATE(),

Qty INT,

CONSTRAINT pid_chk CHECK (pid LIKE '[P|S|T][0-5][0-5][0-5]'),

CONSTRAINT qty_chk CHECK (qty > 0)

);

Which of the following statements are true related to the above definition? (Select one or more)

- A) Consider a product with a quantity (qty) of 10. An update written to deduct 15 from the available quantity in the above table will not be successful.
- B) "T12" is a valid pid that could be stored in the Product table.
- C) "P333" is a valid pid that could be stored in the Product table.
- D) Executing the following insert statement will cause an error: INSERT INTO Product(pid, qty) VALUES ('T000', 50)

• E) After executing the following insert statement in the above table, the row inserted will have two null values: INSERT INTO Product(pid, qty) VALUES ('T000', 50)

**Answer: A, C**

---

**Question 54**

Consider the following table definition:

CREATE TABLE Product (

pid CHAR(4) PRIMARY KEY,

pname VARCHAR(30),

manufactureDate DATETIME DEFAULT GETDATE(),

qty INT,

CONSTRAINT pid_chk CHECK (pid LIKE 'P|S|T[0-5][0-5]|10-5|10-577'),

CONSTRAINT qty_chk CHECK (qty > 0)

);

Which of the following statements are TRUE related to the above definition? (Select one or more)

• A) Consider a product with a quantity (qty) of 10. An update statement written to deduct 15 from the available quantity in the above table will not be successful.

• B) T12 is a valid pid that could be stored in the Product table

• C) P333 is a valid pid that could be stored in the Product table

• D) Executing the following insert statement will cause an error:

**Answer: A**

---

**Question 55**

Which of the following statements are true related to constraints? (Select one or more)

• A) If a subtype participates in a relationship that is the same as the other subtypes, that relationship could be added to the supertype

• B) When an entity belongs to only one subtype in the hierarchy, the relationship is total and disjoint

• C) When an entity instance may be a member of multiple subtypes or it does not have to be a member of a subtype, the specialization is overlapping and total

• D) A bowler and batsman (assume wicket keeper is also a batsman), which are subtypes of a cricketer type, is total and disjoint

**Answer: A**

---

**Question 56**

Which of the following statements are TRUE related to constraints? (Select one or more)

• A) A bowler and batsman (assume wicket keeper is also a batsman), which are subtypes of a cricketer type, is total and disjoint

• B) If a subtype participates in a relationship that is the same as the other subtypes, that relationship could be added to the supertype

• C) When an entity belongs to only one subtype in the hierarchy, the relationship is total and disjoint

• D) When an entity instance may be a member of multiple subtypes or it does not have to be a member of a subtype, the specialization is overlapping and total

**Answer: B**

---

**Question 57**

Consider table definition:

CREATE TABLE Product (

pid CHAR(4) PRIMARY KEY,

pname VARCHAR(30),

manufactureDate DATETIME DEFAULT GETDATE(),

Qty INT,

CONSTRAINT pid_chk CHECK (pid LIKE '[P|S|T][0-5][0-5][0-5]'),

CONSTRAINT qty_chk CHECK (qty > 0)

);

Which statements are true? (Select one or more)

• A) T12 is a valid pid that could be stored in the Product table.

• B) Consider a product with a quantity (qty) of 10. Updating the table to deduct 15 will not be successful.

• C) After executing INSERT INTO Product(pid, qty) VALUES('T000', 50), the row inserted will have two null values.

• D) P333 is a valid pid that could be stored in the Product table.

• E) Executing INSERT INTO Product(pid, qty) VALUES('T000', 50) will cause an error

**Answer: B, D**